Constructor University Bremen

MPFR LAB II

Part 6: Servo (Angle/Position) Control
Spring Semester 2025

Task1.25-Task1.27

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Task 1.25. *Show that under the assumption of negligible L_a,*

$$H_{er}(s) = \frac{s^2 D(s)}{Q(s)}, \tag{1.49d}$$

$$Q(s) = R_a J\, s^3 + s^2(R_a b + k_e k_t + k_t K_D) + k_t K_P\, s + k_t K_I \tag{1.49e}$$

$$H_{av} = \frac{k_t}{s\,D(s)}$$

$$D(s) = (R_a + L_a s)(Js + b) + k_e k_t$$

$$= R_a (Js + b) + k_e k_t$$

$$G(s) = K_p + K_I \frac{1}{s} + K_D s$$

$$\Rightarrow \quad H_{er}(s) = \frac{1}{1 + H_{av}(s)\,G(s)}$$

$$= \frac{1}{1 + \frac{k_t}{s\,D(s)} G(s)}$$

$$= \frac{1}{1 + \frac{k_t}{s\,D(s)} \left(K_p + K_I \frac{1}{s} + K_D s\right) \frac{s}{s}}$$

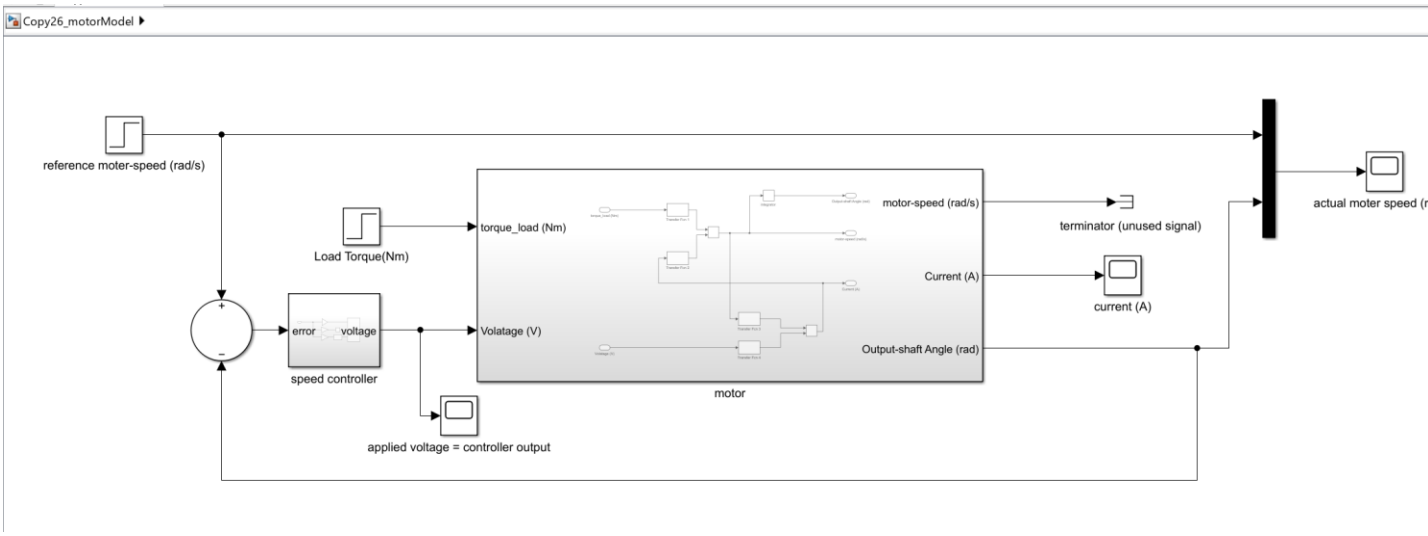
$$= \left(\frac{s^2 D(s) + k_t K_p s + k_t K_I + K_D s^2 k_t}{s^2 D(s)} \right)^{-1}$$

$$= \frac{s^2 (R_a (Js + b) + k_e k_t)}{s^2 (R_a (Js + b) + k_e k_t) + k_t K_p s + k_t K_I + K_D k_t s^2}$$

$$= \frac{s^2 (R_a (Js + b) + k_e k_t)}{s^3 (R_a J) + s^2 (R_a b + k_e k_t + K_D k_t) + s (k_t K_p) + k_t K_I}$$

$$\Rightarrow Q(s)$$

Task 1.26. Copy the model-file for the PID speed-controller to a new file. Now change the speed-controller to a servo-controller as shown in Fig. 1.19.



Task 1.27. Proceed as follows:

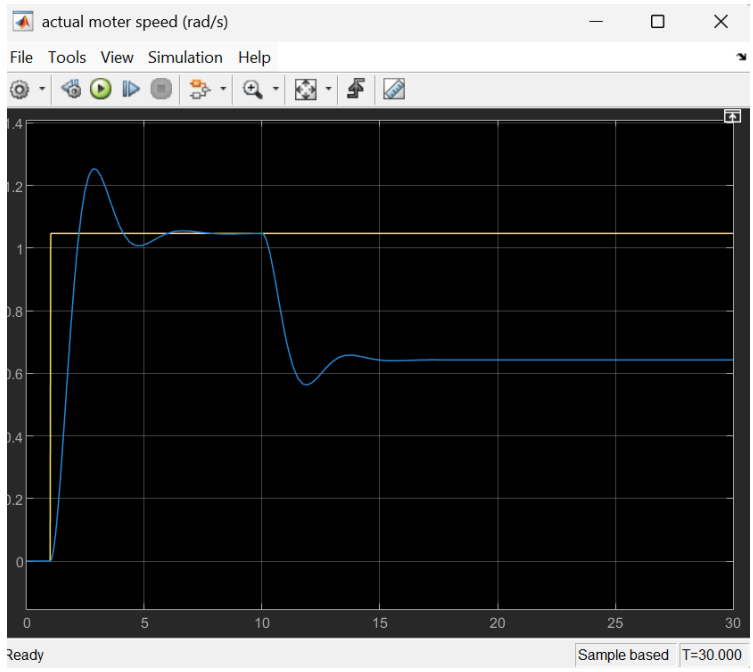
1. Start with some relatively small value of KP such as 1, keeping KD and KI zero. Set the reference input to π/3 rad and TL ≡ 0. Simulate the sytem for 30 seconds. Create a table with the following columns and jot down your observations: KP , KI , KD, max absolute value of current, max absolute value of voltage, max overshoot of the angle above the reference, settle-time (approx. time it takes for the angle to settle near its steady-state value).

2. Now add some damping by setting KD = 0.1. Enter your observations in the table.

3. Now add some integral control KI = 0.1. Enter your observations in the table.

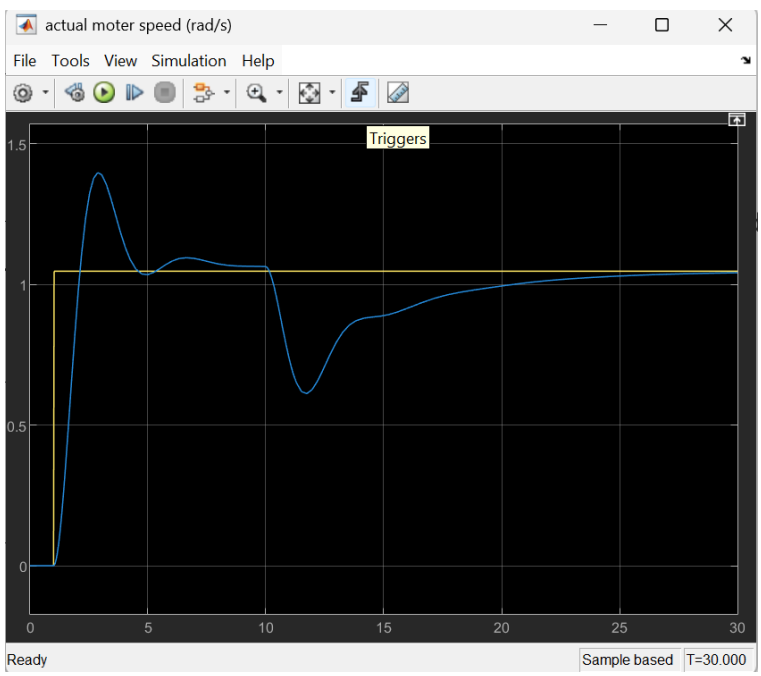
K_P	**K_I**	**K_D**	**Max absolute value of current**	**Max absolute value of voltage**	**Max overshoot of the angle**	**Settle time**
1	0	0	0.436	1.47	1.408	~7.5
Remain unchanged	*Remain unchanged*	0.1	4.696	11.519	1.266	~7.028
Remain unchanged	0.1	*Remain unchanged*	4.696	11.519	1.307	~10

4. Now we will see what the controller can do if we add a step load torque. First reset KI = 0. Reduce KP to 0.5 and KD to 0.001. Make a step load of TL = 0.01 Nm at t = 10 seconds. Did the system reach its reference value?



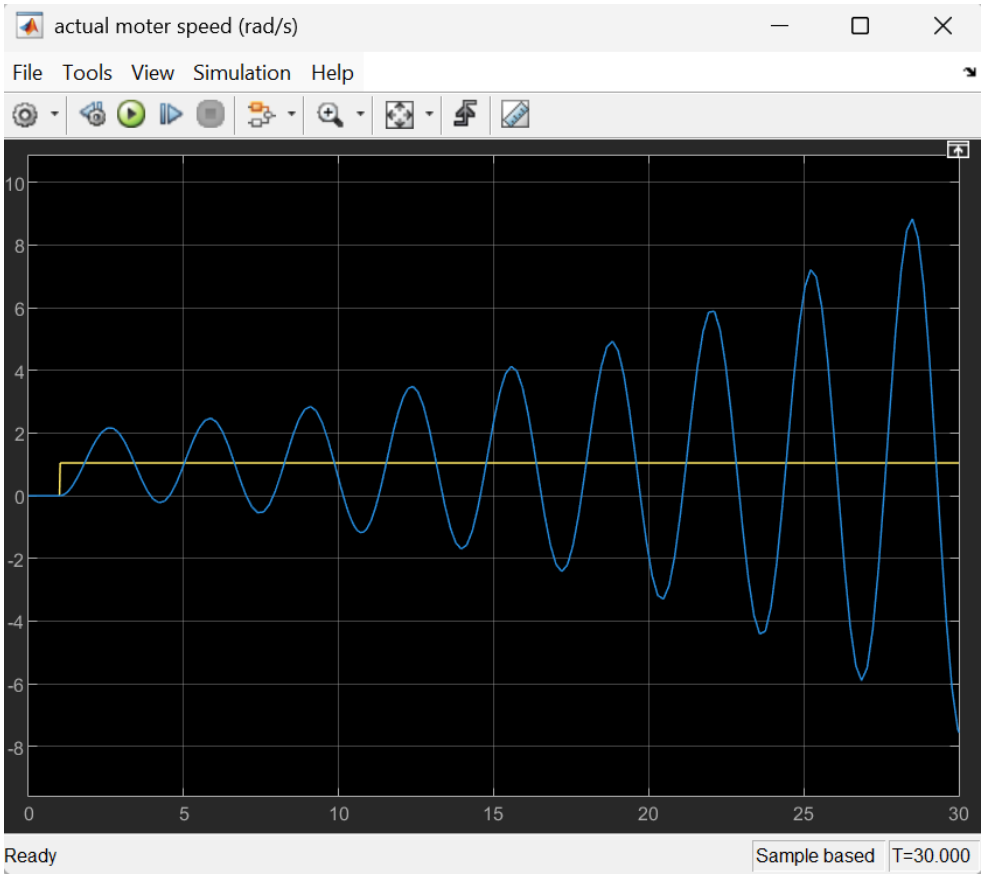
As shown above, the system didn't reach its reference value

5. Now increase KI to 0.1 and simulate again. Do you see any difference in the steady-state error?



As shown above, the system reaches its reference value

6. Although it is now tempting to increase KI even more, we should be careful. Making KI too big can easily make the closed-loop system unstable. Let us see how an unstable system reacts: set KI = 1 and simulate.



7. Does our model also predict this instability? Find the poles of the system by finding the roots of Q(s) for these controller parameters. Are they in LHP?